

STUDENT PERFORMANCE PREDICTION USING MACHINE LEARNING

1. Introduction

Machine Learning helps systems learn from data and make predictions automatically. This project uses a Student Performance Dataset to predict the final student grade (G3) using Linear Regression and Random Forest Regression models.

2. Objective

- Preprocess student data
- Encode categorical variables
- Apply feature scaling
- Train machine learning models
- Predict final student grades
- Compare model performance

3. Dataset Description

The dataset contains student demographic, social, and academic information such as school, gender, age, study time, failures, absences, and grades (G1, G2, G3). The target variable used for prediction is G3 (Final Grade).

4. Tools and Libraries Used

Python, Pandas, NumPy, Matplotlib, and Scikit-learn were used for preprocessing, training, evaluation, and visualization.

5. Methodology

The dataset was loaded using Pandas. Missing values were removed and categorical columns were encoded using LabelEncoder. Feature scaling was performed using StandardScaler before training the machine learning models.

6. Machine Learning Models

- Linear Regression – Predicts values using a linear relationship between variables.
- Random Forest Regressor – Uses multiple decision trees for accurate prediction.

7. Model Evaluation

The models were evaluated using RMSE (Root Mean Squared Error). Lower RMSE values indicate better prediction accuracy. The performances of Linear Regression and Random Forest were compared.

8. Visualization

A scatter plot was created to compare Actual G3 values and Predicted G3 values using the Random Forest model. The graph helps visualize prediction accuracy.

9. Output Generated

- ml_output.csv – Predicted output values
- prediction_plot.png – Actual vs Predicted graph

10. Insights and Conclusion

The project successfully predicted student performance using machine learning algorithms. Random Forest generally performed better than Linear Regression. The analysis shows that study time, attendance, and previous grades influence student performance.

Submitted By:

Amala Binisha C

M.Sc. Data Science